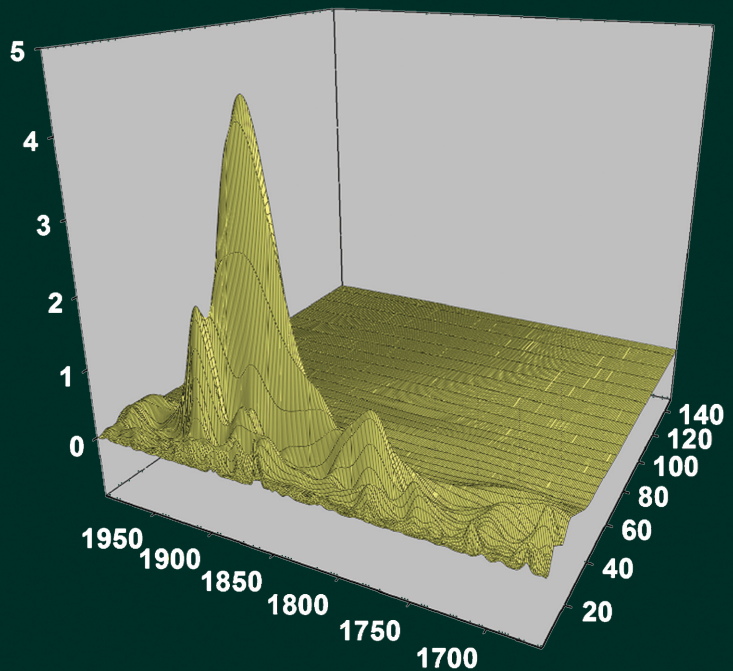
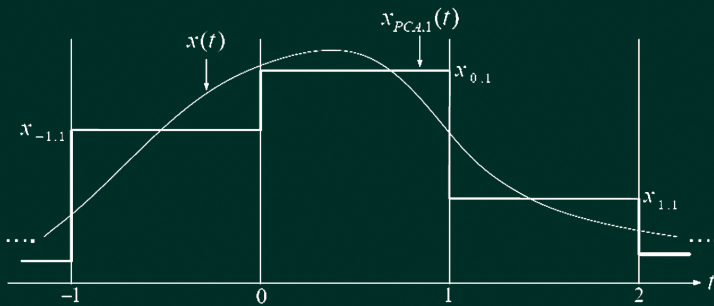


Wavelets and Fractals in Earth System Sciences



Editors

E. Chandrasekhar

V. P. Dimri

V. M. Gadre



CRC Press
Taylor & Francis Group

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Geomagnetic Jerks: A Study Using Complex Wavelets

E. Chandrasekhar, Pothana Prasad, and V. G. Gurijala

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Introduction

The naturally produced geomagnetic field variations recorded at the earth's surface or at satellite altitudes is of dual origin: external and internal. The external part originates from the current systems generated in the ionosphere and the distant magnetosphere by the interaction of the solar wind with them. As a result, different types of geomagnetic variations based on the solar intensity having different periodicities are generated. A small portion of the internal part arises due to the eddy currents induced in the conductive layers of the earth by the time-varying external field diffusing within the earth. In addition to induction by external sources, the major contribution for the internal part, known as the earth's permanent magnetic field or dipole field, originates from the earth's liquid outer core. This is a slowly varying field with time. More than 95% of the earth's magnetic field is believed to

be due to the electric currents generated within the fluid outer core. An additional contribution of approximately 2% comes from the earth's lithosphere and crust. Thus, the science of geomagnetism opens up a plethora of research topics related to terrestrial and extraterrestrial phenomena, and facilitates the studies of external source current systems, secular variations, and electrical conductivity variations within the earth. Figure 7.1 depicts a schematic of the broad classification of geomagnetic field variations. Figure 7.2 describes the vector representation of the earth's magnetic field at any point in the northern hemisphere of the earth.

Figure 7.3a shows an example of long-period geomagnetic field variations, whose first time-derivative, known as geomagnetic secular variations (Figure 7.3b), owe their origin to the convection currents generated by the dynamo processes actively taking place within the earth's liquid outer core. It has been observed that the geomagnetic data of several decades show some sudden changes in the slope of the secular variations (i.e., second time-derivative of the long-period geomagnetic field), having periods of approximately 1 year. Such features are generally known as secular accelerations or geomagnetic jerks (GJs; Figure 7.3c; Courtillot et al. 1978; Courtillot and Le Mouel 1976, 1984; Backus 1983; Gubbins and Tomlinson 1986; Mandea et al. 2000). Barraclough et al. (1989) described the use of horizontal components of the magnetic field to understand the outer core motions. GJs are easily noticeable in the east–west component (Y or D , Figure 7.2) of the magnetic field, although it is sometimes seen in the north–south component (X or H , Figure 7.2) as well (Bloxham et al. 2002). Well-known global jerks have occurred

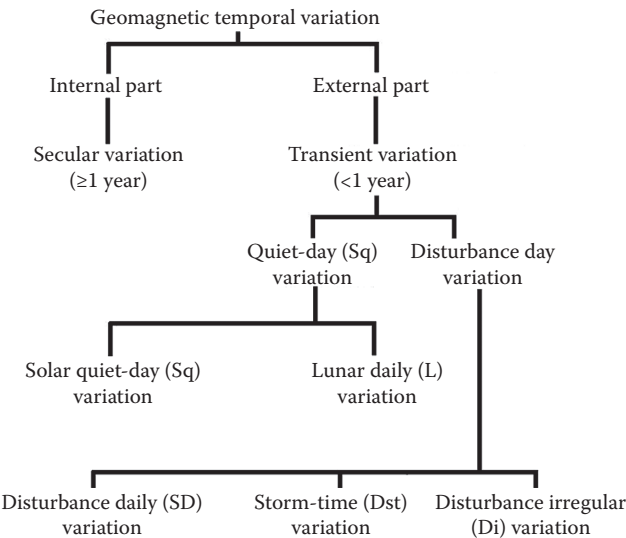


FIGURE 7.1
A schematic showing different types of temporal geomagnetic field variations.

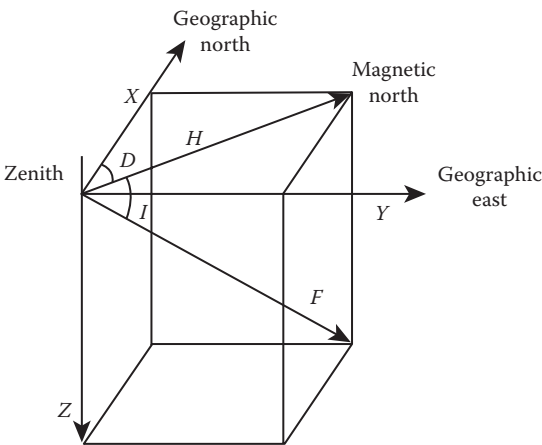


FIGURE 7.2
Elements of the geomagnetic field in northern hemisphere. The declination (D) refers to the angle between geographic north (X) and geomagnetic north (H). The inclination (I) refers to the angle between the geomagnetic north and the total field (F). Y denotes the geographic east component and Z refers to the direction of the vertical magnetic field.

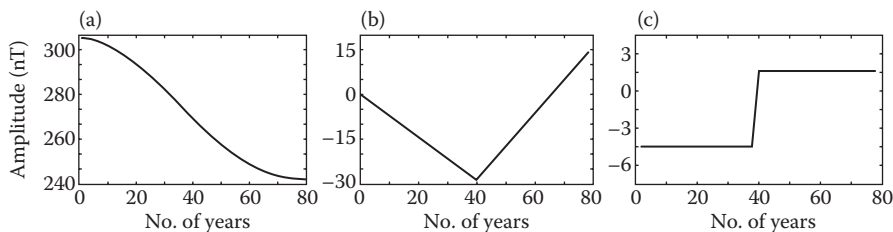


FIGURE 7.3
A schematic for explanation of GJ phenomenon. (a) Example of decadal variations of geomagnetic east–west component, Y ; (b) secular variations, obtained by taking dY/dt ; (c) secular acceleration (d^2Y/dt^2) or GJ, showing a sudden jump at the point of inflection in (a).

during the years 1969 to 1970, 1978 to 1979, and 1991 to 1992 and local jerks during 1901 to 1902, 1913 to 1914, 1925 to 1926, 1932 to 1933, 1942 to 1943, 1949 to 1950, around 1985, and in 1999 to 2000 (Alexandrescu et al. 1995, 1996; De Michelis et al. 1998; De Michelis and Tozzi 2005; Adhikari et al. 2009).

The origin of GJs has been the subject of a lot of debate. Employing a spherical harmonic analysis (SHA) technique, Malin and Hodder (1982) reported that GJs are of internal origin. Later, claiming some loopholes in Malin and Hodder’s analysis, Alldredge (1984) argued that GJs are of external origin. However, further studies by Gavoret et al. (1986), Davis and Whaler (1997), Le Huy et al. (1998), Nagao et al. (2002), and Bloxham et al. (2002) confirmed the internal origin of GJs. Gavoret et al. (1986) separated the internal and external field components using geomagnetic indices and found a negligible

influence of external sources on GJs. Using the time-dependent flow model of Bloxham and Jackson (1992), Davis and Whaler (1997) studied the outer core fluid flow behavior with respect to the 1969 GJ event and showed that GJs cannot be produced by an instantaneous change in the flow. However, a steady flow with one spherical harmonic flow coefficient changing with time can produce a GJ in the secular variation field at the surface of the earth. Computing the spherical harmonic models for the 1969, 1978, and 1991 jerk events, Le Huy et al. (1998) discussed the characteristics of these successive GJs. Nagao et al. (2002) applied statistical time series modeling to the monthly means of geomagnetic data and discussed the noninfluence of external currents (field-aligned current and ring current) on the occurrence of GJs. Bloxham et al. (2002) explained that the GJs arise due to the combination of a steady flow and a simple time-varying, axisymmetric, equatorially symmetric, toroidal zonal flow of the core fluid. They also opined that the short duration of GJs is due to the differential fluid flow at the surface of the earth's outer core. Therefore, based on the theoretical and observational evidence, it is now well established that GJs are of internal origin.

GJs are believed to occur sometimes globally and sometimes locally,* which has led to the belief of the existence of hemispherical differences in their occurrences (Backus 1983). Differences in their spatiotemporal occurrences have led to studies in the role of lower mantle conductivity as one of the reasons for such a diverse nature of occurrences of GJ events (Acache et al. 1980; Ducruix et al. 1980; Backus 1983; Alexandrescu et al. 1999; Nagao et al. 2003).

In the present study, for the first time, we introduce complex wavelets and demonstrate how they would be useful in understanding the phase characteristics of GJs, which in turn are related to the fluid flow directions in the outer core, which is responsible for the generation of GJs. Using real wavelets, we first discuss the presence or absence of GJs in the global ground magnetic observatory data of 75 years from 74 observatories, by evaluating ridge functions, which signify a log-linear relation between the absolute value of wavelet transform coefficients and the wavelet scale (see Alexandrescu et al. 1995). Next, using complex wavelets, we discuss the phase characteristics of GJs (global or local). We follow a systematic approach by making a groupwise study of global magnetic observatory data corresponding to (i) polar regions, (ii) high-latitude to mid-latitude regions, (iii) mid-latitude to equatorial regions, and (iv) southern hemisphere regions. For the first time, we demonstrate that phase information, together with ridge functions, provides an unequivocal confirmation not only about the occurrence of a GJ in the data but also about the changes in the direction of fluid flow at the time of its occurrence. Finally, we discuss the spatiotemporal behavior of GJs in the light of some hemispherical differences observed at the time of their occurrences.

* The terms *global* and *local* are purely relative, and as to how global is "global" and how local is "local" are not well defined.

Role of Wavelets and Complex Wavelets in Geomagnetism

Occurrences of GJs in the decadal secular variation data have been understood to be random phenomena and show not only a varied spatiotemporal behavior (sometimes occurring as a global feature and sometimes as a local one) but also a varied time–frequency localization (i.e., irregular occurrences of these high-frequency GJs in time). As a result, conventional signal processing techniques fail to facilitate a better understanding of the overall characteristics of GJs. Wavelet analysis is an effective mathematical tool to achieve the time–frequency localization of GJs (see the Wavelets in Geophysics section in Chapter 1). Such natural events, having diverse space–time and time–frequency characteristics, can be well understood with wavelet schemes (see for example, Alexandrescu et al. 1995; Adhikari et al. 2009). Another interesting characteristic of GJs is that they occur regardless of the direction in the convective motion of the outer core fluid flow (clockwise or anticlockwise). Such fluid flow directions clearly manifest as phase changes in GJs, which can be well understood with the help of complex wavelets. Complex wavelets provide more relevant information on the phase characteristics of the transients present in the signal than real wavelets do (e.g., Unser and Aldroubi 1996; Haddad et al. 2004; Zhou et al. 2009).

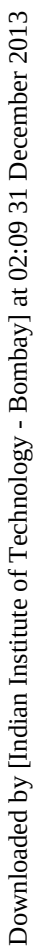
Data Processing of Global Magnetic Observatory Data

The geomagnetic east–west component (D or Y), usually recorded in units of arc minutes at any observatory, were first converted into magnetic field units (nanotesla, nT), using

$$D(\text{nT}) = \left[\frac{\pi}{180} \times \frac{H_{\text{abs}}}{60} \right] \times D(\text{min}) \quad (7.1)$$

H_{abs} refers to the absolute value of H component at that magnetic observatory. Similarly, to convert Y variations at any observatory from the units of arc minutes to nanotesla, substitute X_{abs} of that observatory for H_{abs} in the above equation.

Monthly means of D (or Y) variations were used for the present study. They were estimated using the daily mean values, which in turn were calculated with hourly mean values procured from the world data center, WDC-C2 for Geomagnetism, Kyoto University, Japan, and from WDC at Edinburgh. The global magnetic observatory data considered for the present study correspond to the years 1925 to 2000 (Figure 7.4). At places, wherever a whole month's data were missing, they were interpolated using the average of the previous and



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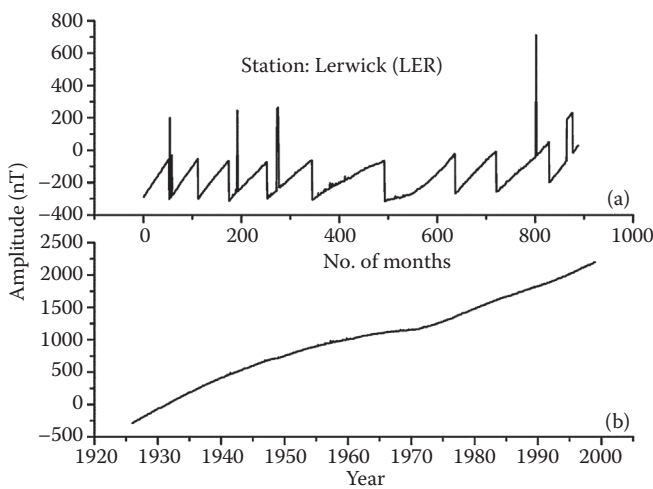


FIGURE 7.5 Raw data (a) and fully processed data (b) of geomagnetic east–west component at one of the high-latitude observatories, LER (see Figure 7.4 and Table 7.1).

Wavelet Analysis

Continuous wavelet transformation (CWT) was performed using real and complex wavelets on the fully processed data of geomagnetic east–west components of approximately 75 years, recorded from 74 global magnetic observatories. Because the occurrences of GJ events are intermittent and sporadic, they can be treated as singularities in the decadal variations of the magnetic field. Accordingly, CWT is ideally suited to detect them. It may be recalled from Chapter 1 that in wavelet transformation, the signal to be transformed is convolved with the mother wavelet and the transformation is computed for different segments of the data at different translations and dilations (see the subsection Wavelets in the Basic Theory and Mathematical Concepts of Wavelets and Fractals in Chapter 1). The wavelet transformation in which the dilations and translations will be continuously varied is called CWT (see the Continuous Wavelet Transformation section in Chapter 1), and the transformation in which they will be varied as power of an integer n (that is, $n!, j = 1, 2, 3, \dots k$) is called discrete wavelet transformation (DWT; see the Discrete Wavelet Transformation section in Chapter 1). Generally, in DWT, dyadic scales and shifts are used, in which case $n = 2$. Although both CWT and DWT are linearly translational-invariant (see the Translational Invariance section in Chapter 1), the difference between them is as follows. In CWT, an amount of small time shift (δt) in the wavelet function results in the same amount of shift in the transformed domain. As a result, abrupt changes in the data can be effectively picked up by performing the computations with continuous shifts. On the other hand, in DWT, the

wavelet transform coefficients become translational-invariant only if the translations are in dyadic steps [that is, as a power of 2; see Ma and Tang (2001)]. Therefore, CWT, in general, is best suited for such studies in which abrupt jumps, discontinuities, spikes, singularities, etc., in the data are of interest for analysis.

CWT with real wavelets was first performed to ensure the presence or absence of GJs in the decadal variations of magnetic data. Next, the CWT with complex wavelets was done to study the phase characteristics of the identified GJs. The presence of a GJ in the data can be identified by examining the linear relation between the logarithm of the absolute value of wavelet transform coefficients and the logarithm of the wavelet scale.

Estimation of Ridge Functions Using Real Wavelets

Because the occurrences of GJ events are causal* in nature, they can be treated as independent events (singularities) and accordingly, each event can be analyzed separately. Furthermore, because they manifest the second time-derivative of the geomagnetic field, these singularities can be represented as some α th derivative of the signal (α being the regularity of the singularity). So, the time-varying magnetic field, say, $f(t)$ can be expressed as (Alexandrescu et al. 1995)

$$f(t) = A(t + \delta t)^\alpha + h(t) + n(t) \quad (7.1)$$

where $h(t)$ is the long-period harmonic component and $n(t)$ is the noise term, as a function of time. The factor $A(t + \delta t)$ with the condition

$$(t + \delta t)^\alpha = \begin{cases} 0 & t \leq \delta t \\ (t + \delta t)^\alpha & t > \delta t \end{cases} \quad (7.2)$$

defines the presence of a singularity in the signal centered around δt , having amplitude A . Because the singularity is included in $f(t)$, the wavelet transformation can detect it and can also tell about the time of its occurrence. Performing wavelet transformation on Equation 7.1 and doing some subsequent simple mathematical calculations (see Alexandrescu et al. 1995), we obtain ridge functions that represent a linear regression between the logarithm of the absolute value of wavelet transform coefficients along the lines of maxima, $\log_2(|CWT_{\tau,s}|)$, and the logarithm of scale, $\log_2(s)$. Mathematically, it is expressed as

* A time signal $f(t)$ is said to be causal if it is zero before the time of its occurrence. Mathematically, it is represented as
$$f(t) = \begin{cases} 0 & t < 0 \\ f(t) & t \geq 0. \end{cases}$$

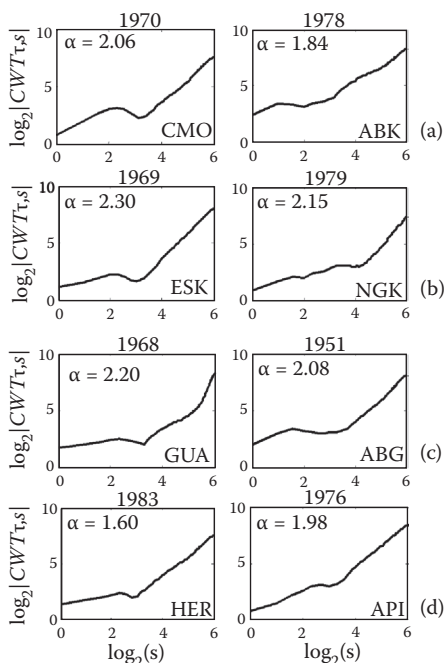


FIGURE 7.6

Ridge function plots of a few stations, each corresponding to polar region (a), high-latitude to mid-latitude region (b), mid-latitude to equatorial region (c), and southern hemisphere (d) stations. The regularity value or the slope, α , is computed by fitting a linear fit to the data beyond the scale $2^{3.5}$.

$$\log_2(|CWT_{\tau,s}|) = \alpha \log_2(s) + \text{constant} \quad (7.3)$$

where $s > 0$ is the dilation (scale) parameter. Because GJs represent the second time-derivative of the magnetic field, the computed value of α (the slope of Equation 7.3) should be equal to or close to 2. For full mathematical details of derivation of Equation 7.3, the reader is referred to Alexandrescu et al. (1995) and the references therein. Among a variety of real wavelets tested, we have found the Gaus3 wavelet as the optimum one for identifying GJs. Figure 7.6 shows the ridge function plots obtained by using Gaus3 wavelet corresponding to GJ events observed at a few stations in the polar region (Figure 7.6a), high-latitude to mid-latitude region (Figure 7.6b), mid-latitude to equatorial region (Figure 7.6c), and southern hemisphere stations (Figure 7.6d). The α value or the slope is computed by fitting a linear fit to the data beyond the scale $2^{3.5}$.

Complex Wavelet Analysis of GJs

The complex continuous wavelet transform (CCWT), like the real wavelet transform, expands a signal in terms of a set of complex wavelet basis functions, $\psi_{\tau,s}^*$, which are dilated (scaled) and translated (shifted) versions of the complex wavelet. The real and imaginary parts of complex mother wavelet, $\psi^*(t) [= \psi_r(t) + i\psi_i(t)]$, individually satisfy all the conditions of the real-valued mother wavelets. The CCWT coefficients, $CCWT_{\tau,s}$, are estimated by

$$CCWT_{\tau,s} = \frac{1}{\sqrt{s}} \int f(t) \psi^* \left(\frac{t-\tau}{s} \right) dt. \quad (7.4)$$

The complex Gaussian wavelet family is derived from the derivative of Gabor wavelet, given as

$$\psi^*(t) = C \frac{d^n}{dt^n} G(t) \quad (7.5)$$

where $G(t) = e^{-(t^2+it)}$ defines the Gabor wavelet, n denotes the order of the derivative, and C is the constant. We have used a variety of complex wavelets, and based on the resolution of the phase information that they had offered, we have identified the complex Gaussian wavelet 1 (Cgau 1, i.e., $n = 1$ in Equation 7.5) as the most suitable one for the present study. From Equation 7.5, it can be easily shown that the real and imaginary parts of the complex Gaussian wavelet form a Hilbert transform pair for all n (Selesnick 2001). As a result, the wavelet transform coefficients, $|CCWT_{\tau,s}|$ become translational invariant (Fernandes 2001; Kingsbury 2001).

Region-Wise Study of GJs

Polar Region

We have analyzed approximately 70 years of data from 11 observatories from this region (Table 7.1). The 1969 to 1970 global jerk event is well resolved in the ABK, ALE, BLC, CMO, LER, and LRV observatories. At all these observatories (except at CMO), the 1969 to 1970 jerk event in the secular variation data is characterized by a convex-shaped curve (slope changing from negative to positive), whereas at CMO, it is characterized by a concave shape (slope changing from positive to negative). An example of phase scalogram plots together with secular variation plots and the ridge function plots related to the 1969 to 1970 event corresponding to

TABLE 7.1
Geographic Coordinates and IAGA Codes of Magnetic Observatories Shown
in Figure 7.4

Sl. No.	IAGA Code	Station Name	Latitude (°N)	Longitude (°E)
<i>Observatories Corresponding to the Polar Region</i>				
1.	ABK	Abisko	68.358	18.823
2.	ALE	Alert	82.497	297.647
3.	BLC	Baker Lake	64.333	263.967
4.	CBB	Cambridge Bay	69.123	254.969
5.	CMO	College	64.87	212.14
6.	HRN	Hornsund	77.0	15.55
7.	IQA	Iqaluit	63.75	291.482
8.	LER	Lerwick	60.133	358.817
9.	LRV	Leirvogur	64.183	338.3
10.	RES	Resolute Bay	74.69	265.105
11.	YKC	Yellowknife	62.482	245.518
<i>Observatories Corresponding to the High-Latitude to Mid-Latitude Regions</i>				
12.	AQU	L'Aquila	42.383	13.317
13.	ARS	Arti	56.433	58.567
14.	BDV	Budkov	49.08	14.015
15.	BEL	Belsk	51.837	20.792
16.	BOU	Boulder	40.14	254.767
17.	BOX	Borok	58.033	38.972
18.	BSL	Stennis Space Centre	30.35	270.36
19.	ESK	Eskdalemuir	55.317	356.8
20.	FCC	Fort Churchill	58.786	265.912
21.	FRD	Fredericksburg	38.21	282.633
22.	FRN	Fresno	37.09	240.283
23.	FUR	Furstenfeldbruck	48.17	11.28
24.	HAD	Hartland	51	355.517
25.	HTY	Hatizyo	33.073	139.825
26.	IRT	Patrony	52.167	104.45
27.	KAK	Kakioka	36.232	140.186
28.	KNY	Kanoya	31.424	130.88
29.	KNZ	Kanozan	35.256	139.956
30.	LVV	Lvov	49.9	23.75
31.	MEA	Meanook	54.615	246.653
32.	MMB	Memambetsu	43.91	144.189
33.	MOS	Krasnaya Pakhra	55.467	37.317
34.	NEW	Newport	48.27	242.883
35.	NGK	Niemegk	52.072	12.675
36.	NVS	Novosibirsk	54.85	83.23
37.	OTT	Ottawa	45.403	284.448

(continued)

TABLE 7.1 (Continued)

Geographic Coordinates and IAGA Codes of Magnetic Observatories Shown in Figure 7.4

Sl. No.	IAGA Code	Station Name	Latitude (°N)	Longitude (°E)
38.	PAG	Panagjurishte	47.485	24.177
39.	PBQ	Poste-de-la-Baleine	55.277	282.255
40.	PET	Paratunka	52.971	158.248
41.	RSV	Rude Skov	55.85	12.45
42.	SIT	Sitka	57.067	224.67
43.	SPT	San Pablo-Toledo	39.547	355.651
44.	SSH	Sheshan (Zo-Se)	31.097	121.187
45.	STJ	Saint Johns	47.595	307.323
46.	TFS	Dusheti (Tblisi)	42.092	44.705
47.	TUC	Tucson	32.17	249.27
48.	VAL	Valentia	51.933	349.75
49.	VIC	Victoria	48.517	236.583
50.	WNG	Wingst	53.743	9.073
<i>Observatories Corresponding to the Mid-Latitude to Equatorial Regions</i>				
51.	ABG	Alibag	18.638	72.872
52.	ANN	Annamalainagar	11.367	79.683
53.	BNG	Bangui	4.333	18.567
54.	DLR	Del Rio	29.49	259.083
55.	GUA	Guam	13.59	144.87
56.	LNP	Lunping	25.0	121.167
57.	HON	Honolulu	21.32	202
58.	MBO	MBour	14.384	343.033
59.	SJC	Cayey, PR	18.11	293.85
60.	UJJ	Ujjain	23.18	75.78
<i>Observatories Corresponding to the Southern Hemisphere</i>				
61.	AIA	Faraday Islands	-65.245	295.742
62.	API	Apia	-13.815	188.219
63.	ASC	Ascension Island	-7.949	345.624
64.	CNB	Canberra	-35.32	149.36
65.	CZT	Port Alfred	-46.431	51.867
66.	DRV	Dumont d'Urville	-66.667	140.007
67.	DVS	Davis	-68.583	77.967
68.	EYR	Eyrewell	-43.422	172.355
69.	GNA	Gnangara	-31.8	116
70.	HER	Hermanus	-34.425	19.225
71.	HUA	Huancayo	-12.05	284.67
72.	PPT	Pamatai	-17.566	210.416
73.	SBA	Scott Base	-77.85	166.763
74.	TSU	Tsumeb	-19.202	17.584

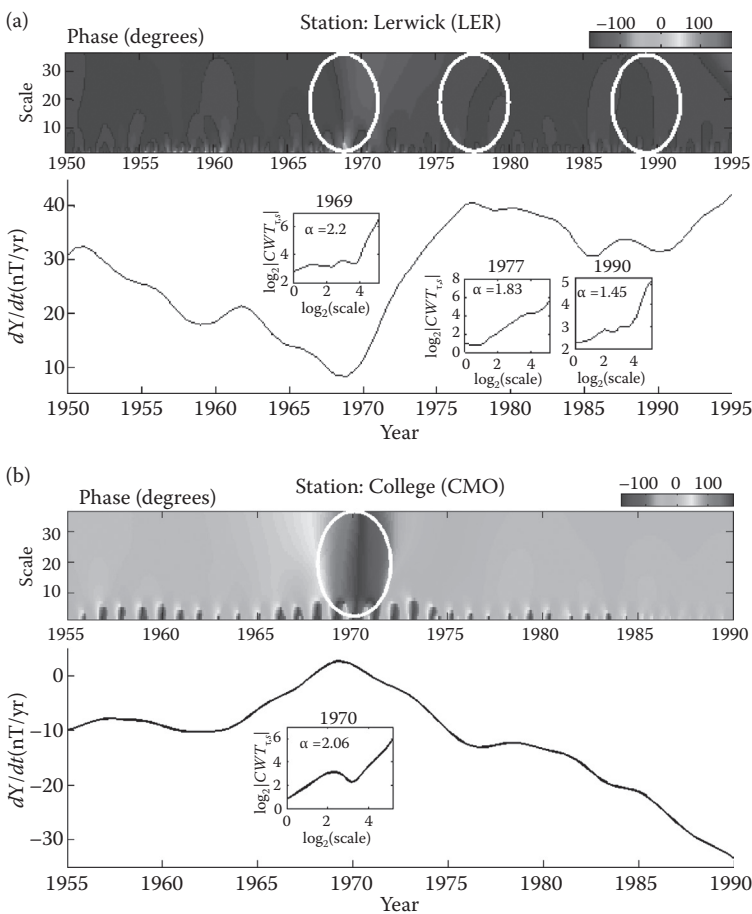


FIGURE 7.7
(See color insert.) Phase scalogram plots together with secular variation plots and the ridge function plots related to the 1969–1970 event corresponding to (a) LER and (b) CMO observatories of polar region.

the LER and CMO observatories are shown in Figure 7.7. Although LER (Figure 7.7a) shows all three global jerk events, CMO (Figure 7.7b) clearly shows only the well-resolved 1969 to 1970 event. However, a study of the phase scalogram plots of all the observatories of this region reveals that the occurrences of all the above three global jerk events were sporadic and not seen at most of the observatories in this region.

High-Latitude to Mid-Latitude Region

We have analyzed approximately 75 years of data from as many as 39 observatories from this region (Table 7.1), since most observatories are located in

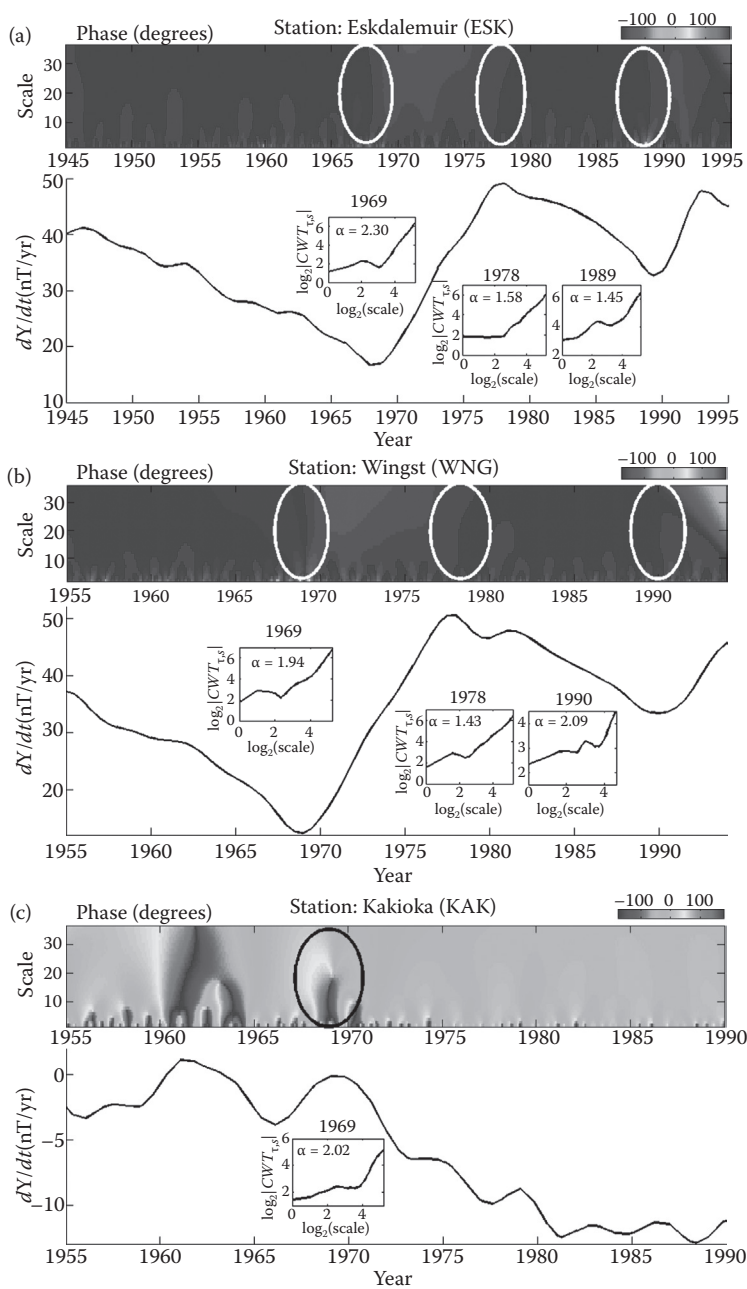


FIGURE 7.8
(See color insert.) Phase scalogram plots together with secular variation plots and the ridge function plots related to the global jerk events corresponding to (a) ESK, (b) WNG, and (c) KAK observatories of the high-latitude to mid-latitude regions.

this region. Figure 7.8 shows the secular variation, phase scalogram plots, and the associated ridge functions corresponding to the three global jerk events of ESK (Figure 7.8a), WNG (Figure 7.8b), and KAK (Figure 7.8c) observatories. At KAK and other Japanese observatories, except for the 1969 event, the other two events could not be well resolved (Figure 7.8c). In this region, the three global jerk events—1969 to 1970, 1978 to 1979, and 1991 to 1992—do not show any differences in the phases corresponding to the same events observed at other observatories, and their corresponding ridge functions also showed a near log-linear nature at these events.

Mid-Latitude to Equatorial Region

We have analyzed about 70 years of data from 10 observatories from this region (Table 7.1). Figure 7.9 shows the secular variation, phase scalogram plots, and the associated ridge functions corresponding to the global as well as local jerk events, as observed at the ABG (Figure 7.9a), BNG (Figure 7.9b), and MBO (Figure 7.9c) observatories. ABG does not show any global events, but shows two clear local jerk events, which occurred during 1950 to 1951 and 1984 to 1985 (Figure 7.9a). In the secular variation plot of BNG (Figure 7.9b), the well-known global GJ event of 1969 showing a concave shape is immediately followed by a well-defined convex shape at 1971. It is intriguing to observe two well defined jerk events within a gap of two years. A similar observation of change of phase is also clearly seen in 1983 (phase changing from positive to negative) and 1990 (phase changing from negative to positive) in a gap of seven years. At MBO (Figure 7.9c), well-defined GJ events showing clear phase changes are seen at the years 1967 and 1982. At almost all observatories in this region, the ridge functions corresponding to global jerk events showed a near log-linear nature and also there are no appreciable phase differences corresponding to the respective events observed at other observatories.

Southern Hemisphere Region

Data from only 14 observatories were available for analysis corresponding to this region (Table 7.1). Figure 7.10 shows the secular variation, phase scalogram plots, and the associated ridge functions corresponding to the global as well as local jerk events, as observed at the HER (Figure 7.10a), PPT (Figure 7.10b), and API (Figure 7.10c) observatories.

Discussion

The GJs that arise due to the differential fluid flow in the outer core are clearly identified by the changes in the slopes of secular variations. They show up as

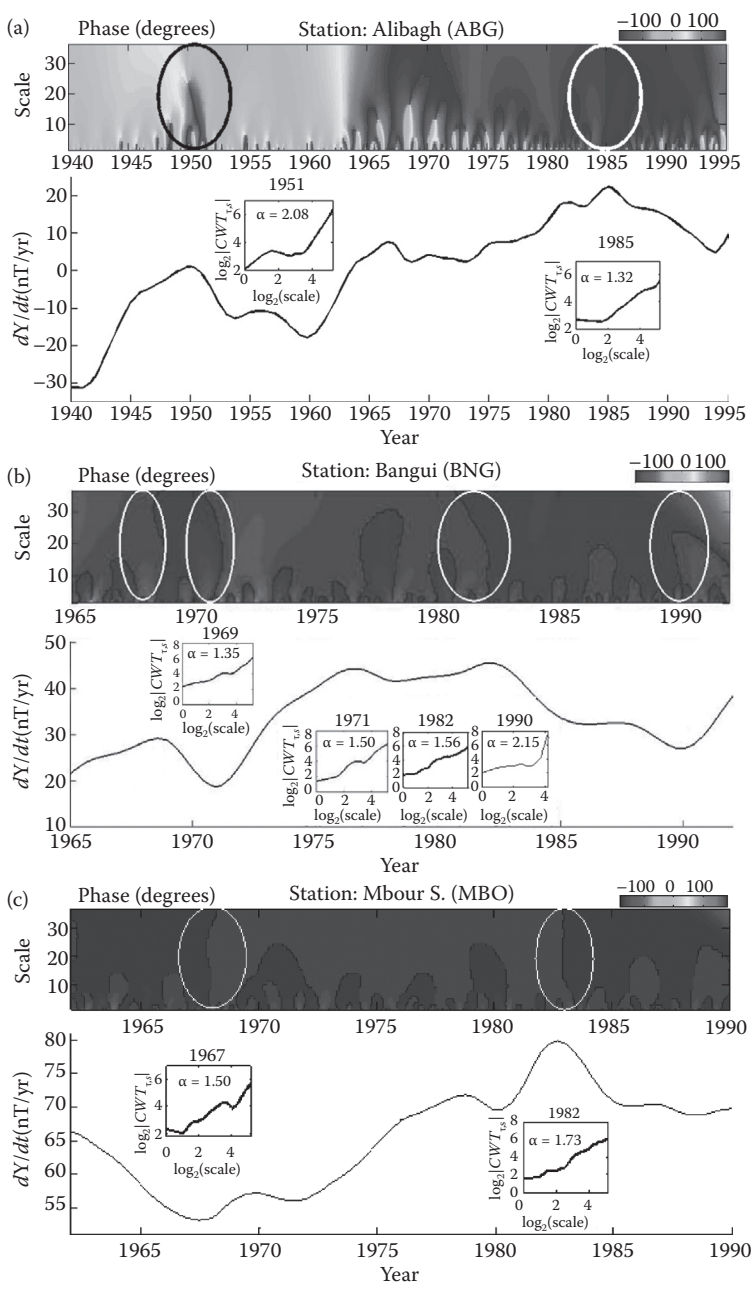


FIGURE 7.9
(See color insert.) Phase scalogram plots together with secular variation plots and the ridge function plots related to the global and local jerk events corresponding to (a) ABG, (b) BNG, and (c) MBO observatories of the mid-latitude to equatorial regions.

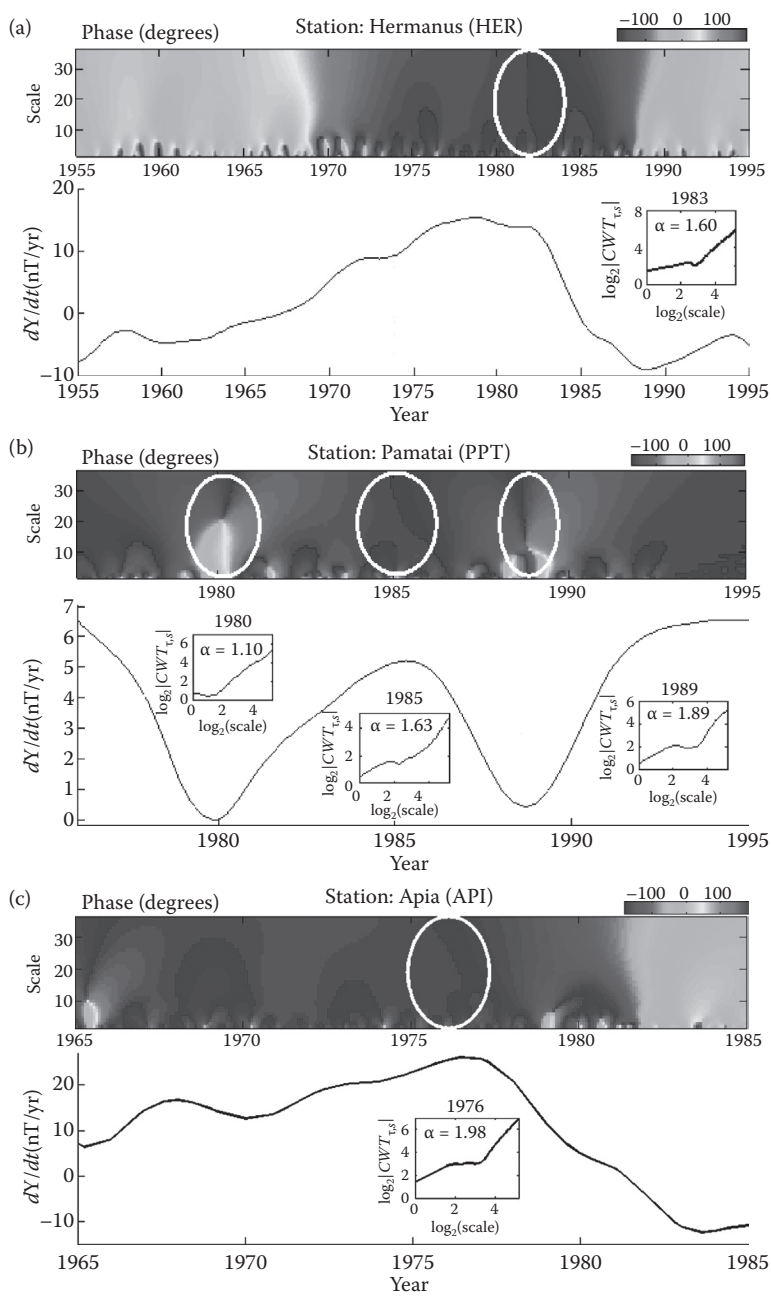


FIGURE 7.10
(See color insert.) Phase scalogram plots together with secular variation plots and the ridge function plots related to the local jerk events corresponding to (a) HER, (b) PPT, and (c) API observatories of the southern hemisphere region.

convex-shaped or concave-shaped curves in the secular variation data, whose extremum (local maximum or minimum) signifies the time of occurrence of jerks. In this study, an analysis of GJs was done using real and complex wavelets. Although the former were used to first ensure the presence or absence of GJs in the data, the latter were used to understand the phase characteristics of the identified GJs. As explained earlier, the linear regression between the logarithm of wavelet transform coefficients and the logarithm of wavelet scale (i.e., the ridge function) confirms the presence or absence of GJs in the data. Figure 7.6 shows the ridge function plots depicting the occurrence of some global as well as some local GJs at some select observatories corresponding to different regions on the globe. The ridge functions associated with jerk events are linear at scales $>2^{3.5}$ (~12). Because the monthly mean values were used in the analysis, perhaps this number is related to the 1-year period associated with jerk events. The portions of the ridge function curves at scales $<2^{3.5}$, which are not linear, are associated with extraterrestrial noise (Alexandrescu et al. 1995) affecting the data. The scale values in the range $2^{3.5}$ to 2^4 mark the transition between the part of the curve dominated by external noise and the part dominated by GJs. If the influence of external noise is high in the data, then the identification of time of occurrence of the jerk event will be less accurate. However, at some observatories, for a couple of ridge functions, the linear nature of the curves starts at scale values less than $2^{3.5}$ [see, for example, the 1978 event at ESK (Figure 7.8a) and the 1969 and 1978 events at WNG (Figure 7.8b)]. This possibly suggests that the wavelet may be averaging out the external noise during the CWT operation. At some stations, the linear nature of the curves starting at higher scales (i.e., $s > 2^4$) is less perfect, in the sense that some kinks are observed in the linear portion of the curves. It is not clear, if a strong long-period external noise in the data is responsible for this. Furthermore, it is also important to note that the α values estimated from Equation 7.3 are not the intrinsic property of the jerk alone, but also depend on the wavelet used (Adhikari et al. 2009).

Haddad et al. (2004) show that the phase scalogram plots corresponding to the complex wavelet transform of synthetic data suggest a clear phase change from $-\pi$ to $+\pi$ (corresponding to convexity) and $+\pi$ to $-\pi$ (corresponding to concavity) at different discontinuities present in the data. In case of GJ phenomena, such phase changes imply changes in the direction of large-scale convective motions on the surface of the outer core. A careful observation of phase scalogram plots together with respective secular variation data and the associated ridge functions corresponding to the global and local jerks shown in Figures 7.7 through 7.10 provide some useful clues for better understanding the nature of the fluid flow at the time of occurrence of different GJs.

The European observatories show a clear signature of the occurrence of global jerks (Figure 7.8a and b). The near log-linear nature of the ridge functions corresponding to these jerks at most European observatories also supports this. Furthermore, there is also a clear change in the phase of these

discontinuities from $-\pi$ to $+\pi$ or $+\pi$ to $-\pi$ at the time of the occurrence of GJs. A careful examination of Figures 7.7a and 7.8b shows a clear double peak in the secular variation data around the time of the 1978 to 1979 jerk event at a high-latitude station (LER) and at a European station (WNG), respectively. Around the same time, another European observatory, Niemegk (NGK), also observed such a feature (not shown here). This phenomenon is understood to be a manifestation of time-varying outer core fluid flow (Bloxham et al. 2002). At most of the observatories, while the 1969–1970 global event shows a clear phase change from $-\pi$ to $+\pi$, the corresponding phase changes for the same events at CMO (Figure 7.7b) and KAK (Figure 7.8c), are from $+\pi$ to $-\pi$. It is quite intriguing to know what could have caused such an oppositely directed secular acceleration at these two northern hemisphere stations? Secondly, it is also interesting and important to understand, what drives these secular accelerations to reverse their directions within a span of about two decades (the three jerks: 1969, 1978, and 1991 occurred in about two decades), whereas such signatures are not seen at other observatories located in the same hemisphere. Perhaps knowledge on the spatial scale lengths of the outer core convective motions would be quite helpful. Figure 7.11 shows a summary of worldwide distribution of phase changes associated with the three global jerk events of 1969 to 1970 (Figure 7.11a), 1978 to 1979 (Figure 7.11b), and 1991 to 1992 (Figure 7.11c). They show a clear understanding of the nature of outer core fluid motions associated with these jerk events. Figures 7.11a to 7.11c also summarize the hemispherical differences not only in the occurrences of global GJ events but also in their phase characteristics. The poor and sparse distribution of available data in the southern hemisphere also suggests an urgent need to fill such a wide gap in establishing more observatories, so as to have an improved understanding of the GJ phenomena and their associated outer core fluid motions in a broader perspective.

One can also easily observe that the secular variation plots corresponding to most of the observatories show changes in slopes at several years other than the global or local jerk years (Figures 7.7–7.10). The corresponding phase transitions at these years are also notable, particularly only at smaller scales. However, their associated ridge functions do not show the log-linear nature of the curves at higher scales at these years. Perhaps the acceleration in the outer core fluid was not strong enough to drive a GJ.

In addition to the global jerks, the characteristic features seen in secular variations during the years 1951 at ABG (Figure 7.9a), 1983 at HER (Figure 7.10a), 1980 thru 1989 at PPT (Figure 7.10b) and 1976 at API (Figure 7.10c) can be classified as local jerks. They show clear phase changes observed in the complex wavelet phase scalogram plots together with good log-linear nature of ridge functions at these years. This clearly warrants that they be identified as local jerks. Figure 7.9a shows clear change in slope of secular variation data around 1951 at ABG. However, the resolution of the change in phase corresponding to this jerk event in the phase scalogram plot is somewhat poor. We attribute this to the noise associated with this coastal station data.

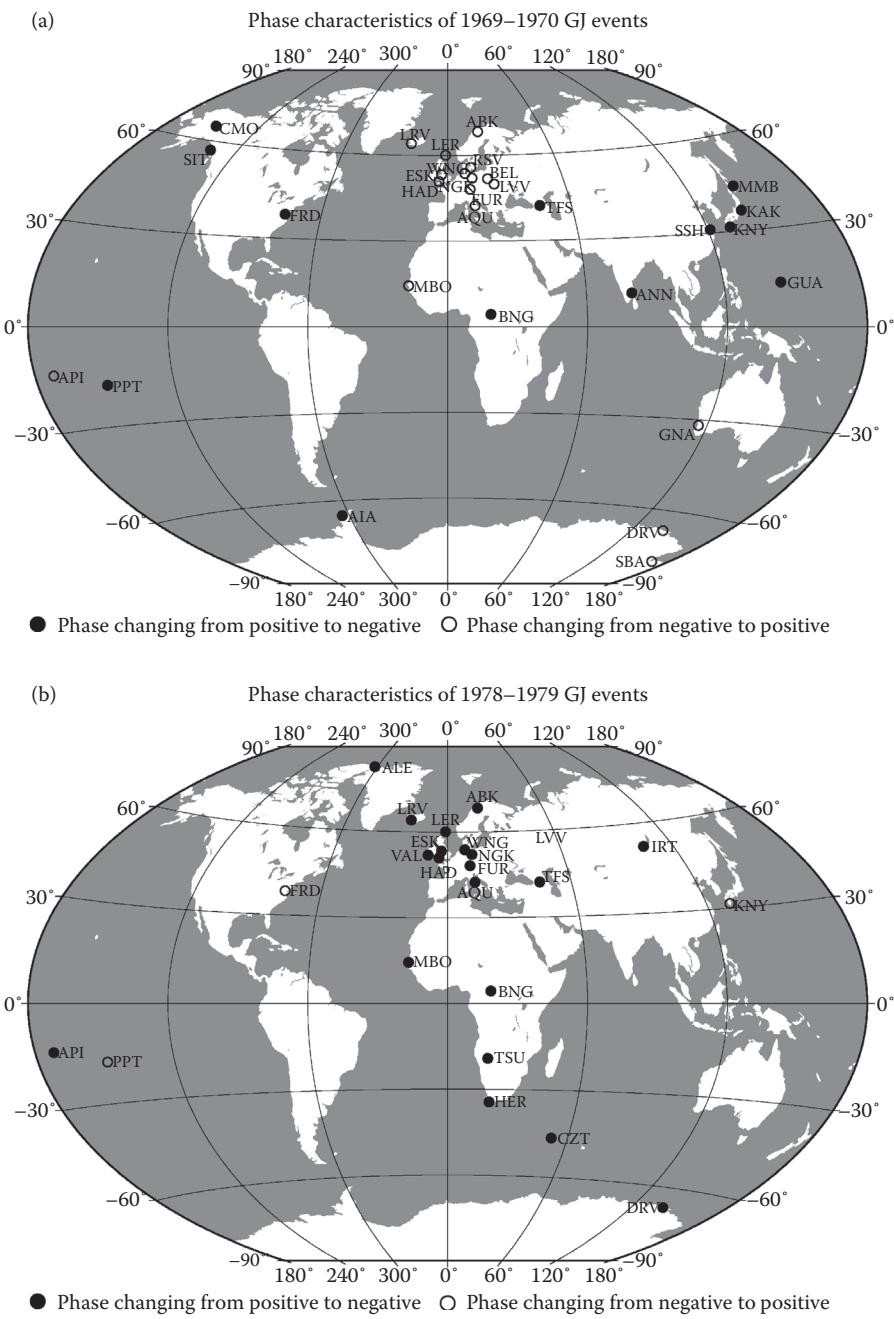


FIGURE 7.11
A summary of worldwide distribution of phase changes associated with the three global jerk events of 1969 to 1970 (a), 1978 to 1979 (b), and 1991 to 1992 (c).

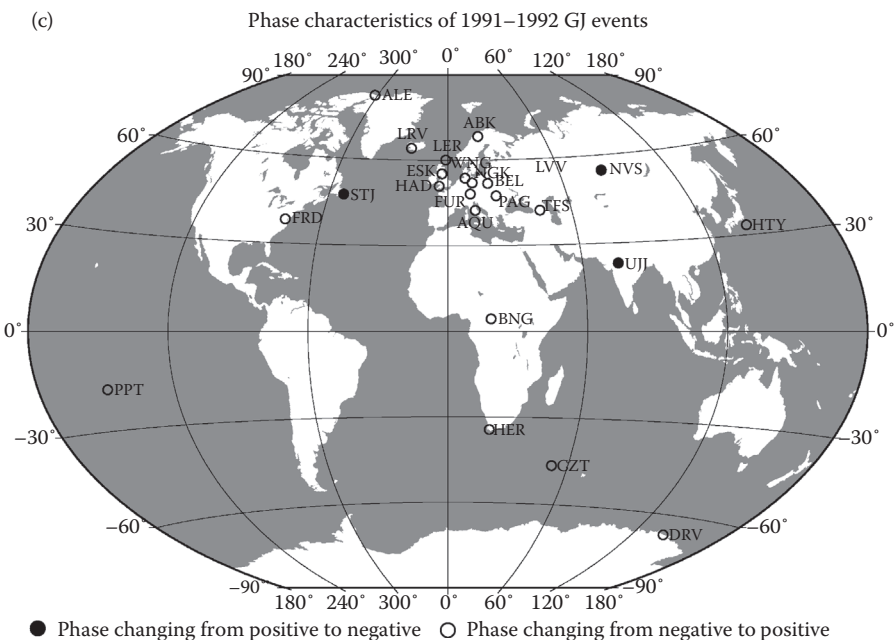


FIGURE 7.11 (Continued)
A summary of worldwide distribution of phase changes associated with the three global jerk events of 1969 to 1970 (a), 1978 to 1979 (b), and 1991 to 1992 (c).

Such a poor resolution in phase change is also seen at another coastal station, KAK (see Figure 7.8c corresponding to the 1969 jerk event). Further, the shape of the secular variation curve associated with the 1951 local jerk at ABG (Figure 7.9a) and the 1969 global jerk at KAK (Figure 7.8c) is rather broad and, accordingly, the phase change is not sharp at the occurrence of this jerk but gradual.

A careful observation of phase information of all the global GJ events summarized in Figure 7.11 deduced from the complex wavelet transformation, has provided unequivocal evidence that there exists some hemispherical differences in the occurrences of global jerks.

We conclude that complex wavelets have helped to understand the nature of outer core fluid motions associated with global as well as local GJs. The hemispherical differences and the delays observed in the time of occurrence of GJ phenomena can be associated with either poor distribution of observatories in the southern hemisphere or a mantle filtering effect as advocated by Backus (1983). Alexandrescu et al. (1999) and Nagao et al. (2003) also relate the delays in observing the jerk features to variations in mantle conductivity. However, this enigma can be solved if more ground magnetic observatories are established in the southern hemisphere and continuously monitored for many years. It is also clearly evident from this study, further to ridge function

plots of GJ events, phase scalogram plots also help in improving the overall understanding of the morphology of global and local GJ events.

The following illustrate some unanswered questions on this topic, in which wavelets can play a significant role in providing a useful and complementary class of information in their effective understanding.

- (i) Why are GJ events clearly seen only in the east–west component of the magnetic field?
- (ii) What drives the convection in the outer core to be global at some instances and local at other times?
- (iii) What makes the hemispherical differences in the occurrences of GJs? Are these time delays solely attributed to lower mantle conductivity variations alone or are any other dynamics in place? Will knowledge of spatial scale lengths of the outer core convective motions be helpful in understanding this?
- (iv) What are the implications of GJs on lower mantle conductivity and vice versa? A further detailed study on better understanding of the morphology of the GJ phenomena, which would provide some answers to the above questions, is important and necessary.
- (v) Finally, are GJs predictable?

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References

- Acache, J., Courtillot, V., Ducruix, J., and Le-Mouél, J.L. 1980. The late 1960's secular variation impulse: Further constraints on deep mantle conductivity. *Physics of Earth Planetary Interiors* 23, 72–75.

- Adhikari, S., Chandrasekhar, E., Rao, V.E., and Pandey, V.K. 2009. On the wavelet analysis of geomagnetic jerks of Alibag magnetic observatory data, India. *Proceedings of the XIII IAGA Workshop on Geomagnetic Observatory Instruments, Data Acquisition and Processing*, edited by Jeffery J. Love, USGS open-file report 2009-1226, 14–23.
- Alexandrescu, M., Gilbert, D., Hulot, G., Le Mouël, J.-L., and Saracco, G. 1995. Detection of geomagnetic jerks using wavelet analysis. *Journal of Geophysical Research* 100, 12557–12572.
- Alexandrescu, M., Gilbert, D., Hulot, G., Le Mouël, J.-L., and Saracco, G. 1996. Worldwide wavelet analysis of geomagnetic jerks. *Journal of Geophysical Research* 101(B10), 21975–21994.
- Alexandrescu, M., Gilbert, D., Le Mouël, J.-L., Hulot, G., and Saracco, G. 1999. An estimate of average lower mantle conductivity by wavelet analysis of geomagnetic jerks. *Journal of Geophysical Research* 104, 17735–17746.
- Allredge, L.R. 1984. A discussion of impulses and jerks in the geomagnetic field. *Journal of Geophysical Research* 89, 4403–4412.
- Backus, G.E. 1983. Application of mantle filter theory to the magnetic jerk of 1969. *Geophysical Journal of the Royal Astronomical Society* 74, 713–746.
- Barracough, D., Gubbins, D., and Kerridge, D. 1989. On the use of horizontal components of magnetic field in determining core motions. *Geophysical Journal International* 98, 293–299.
- Bloxham, J., and Jackson, A. 1992. Time-dependent mapping of the magnetic field at the core-mantle boundary. *Journal of Geophysical Research* 97(B13), 19537–19563. doi: 10.1029/92JB01591.
- Bloxham, J., Zatman, S., and Dumberry, M. 2002. The origin of geomagnetic jerks. *Nature* 420(6911), 65–68.
- Courtilot, V., and Le-Mouël, J.-L. 1984. Geomagnetic secular variation impulses. *Nature* 311, 709–716.
- Courtilot, V. and Le Mouël, J.-L. 1976. On the long-period variations of the Earth's magnetic field from 2 months to 20 years. *Journal of Geophysical Research* 81, 2941–2950.
- Courtilot, V., Ducruix, J., and Le Mouël, J.-L. 1978. Sur une accélération récente de la variation séculaire du champ magnétique terrestre. *Comptes Rendus de l'Académie des Sciences* 287, Série D, 1095–1098.
- Davis, R.G., and Whaler, K. 1997. The 1969 geomagnetic impulse and spin-up of the Earth's liquid core. *Physics of the Earth and Planetary Interiors* 103, 181–194.
- De Michelis, P., Cafarella, L., and Meloni, A. 1998. Worldwide character of the 1991 geomagnetic jerk. *Earth and Planetary Science Letters* 25(3), 377–380.
- De Michelis, P., and Tozzi, R. 2005. A Local Intermittency Measure (LIM) approach to the detection of geomagnetic jerks. *Earth Planetary Science Letters* 235, 261–272.
- Ducruix, J., Courtilot, V., and Le-Mouel, J.L. 1980. The late 1960's secular variation impulse, the eleven year magnetic variation and the electrical conductivity of the deep mantle. *Geophysical Journal of the Royal Astronomical Society* 61, 73–94.
- Fernandes, F.C. 2001. Directional, Shift-Insensitive, Complex Wavelet Transforms with Controllable Redundancy. PhD Thesis, Rice University.
- Gavoret, J., Gilbert, D., Menvielle, M., and Le Mouël, J.-L. 1986. Long-term variations of the external and internal components of the Earth's magnetic field. *Journal of Geophysical Research* 91, 4787–4796.

- Gubbins, D., and Tomlinson, L. 1986. Secular variation from monthly means from Apia and Amberley magnetic observatories. *Geophysical Journal of the Royal Astronomical Society* 86, 603–616.
- Haddad, S.A.P., Karel, J.M.H., Peeters, R.L.M., Westra, R.L., Serdijn, W.A. 2004. Complex wavelet transform for analog signal processing. *Proceedings ProRISC, Veldhoven, November, 2004*.
- Kingsbury, N.G. 2001. Complex wavelets for shift-invariant analysis and filtering of signals. *Journal of Applied and Computational Harmonic Analysis* 10(3), 234–253.
- Le Huy, M., Alexandrescu, M., Hulot, G., and Le Mouél, J.-L. 1998. On the characteristics of successive geomagnetic jerks. *Earth, Planets and Space* 50, 723–732.
- Ma, K., and Tang, X. 2001. Translation-invariant face feature estimation using discrete wavelet transform, In *Wavelet Analysis and its Applications*, Y.Y. Tang et al. (Eds.), LNCS 2251, 200–210. Springer-Verlag, Berlin, Heidelberg.
- Malin, S.R.C., and Hodder, B.M. 1982. Was the 1970 geomagnetic jerk of internal or external origin? *Nature* 296, 726–728.
- Mandea, M., Bellanger, E., and Le Mouél, J.-L. 2000. A geomagnetic jerk for the end of the 20th century? *Earth Planetary Science Letters*, 183, 369–373.
- Nagao, H., Iyemori, T., Higuchi, T., Nakano, S., and Araki, T. 2002. Local time features of geomagnetic jerks. *Earth, Planets and Space* 54, 119–131.
- Nagao, H., Iyemori, T., Higuchi, T., and Araki, T. 2003. Lower mantle conductivity anomalies estimated from geomagnetic jerks. *Journal of Geophysical Research* 108(B5), 2254.
- Selesnick, I.W. 2001. The design of Hilbert transform pairs of wavelet bases via the flat delay filter. *Acoustics, Speech, and Signal Processing* 6, 3673–3676.
- Unser, M., and Aldroubi, A. 1996. A review of wavelets in biomedical applications. *Proceedings of the IEEE* 84(4), 626–638.
- Wessel, P., and Smith, W.H.F. 1995. New version of the generic mapping tools released. *EOS, Transactions of the American Geophysical Union* 76, 329.
- Zhou, G.R., Wang, W.J., and Sun, S.X. 2009. Phase properties of complex wavelet coefficients. *Fourth International Conference on Innovative Computing, Information and Control*. 978-0-7695-3873-0/09, IEEE.